

JIYA SINGHAL

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EXPERIENCE

Sing One Song

Aug 2025 – Present

Software Engineering Intern

Bangalore, India

- **Production Voice Pipeline:** Built the voice-onboarding pipeline from scratch (**FastAPI** + Firebase orchestration, retry-safe Cloud Tasks); re-architected to concurrent orchestration on **GCP Cloud Run** via **asyncio.gather()** + Silero VAD silence-trim, cutting p50 latency **57s → 15s (74%)** on the flow every new user hits first.
- **Data Quality Filter:** Designed a real-time multi-metric audio-quality classifier (SNR + spectral entropy + A-weighted dB + VAD) separating clean speech from unusable recordings, cutting false-positive warnings **5.5% → 0.6%** across 355 production recordings – raising input quality for every downstream model.
- **Benchmarking at Scale (21,750 tests):** Built a **Playwright**-driven harness benchmarking pitch-model accuracy on a synthetic eval set (30 singers × 150 MUSAN noise files × 5 SNR levels), comparing PESTO vs. Aubio and validating a **389KB swift-f0** migration (**97.5% size cut, 98.8% pass**); kept as a standing regression gate.
- **Game Performance & Input Latency:** Diagnosed and fixed input lag across the voice-driven games – a long-standing complaint that singing didn't register instantly – tracing it to serialized pitch inference over an oversized analysis window; shipped behind URL-toggled **A/B variants** with a p50/p95 probe, scoped so the **9 other games** on that shared runtime stayed unaffected.
- **Cross-Platform Audio Reliability:** Root-caused mic failures spanning the **Flutter/WebView** boundary on iOS and Android – a stale **AVAudioSession** category surviving WebView teardown, and an Android audio-focus grab never abandoned on dispose; ruled out four competing hypotheses with source-level proof, then shipped session-repair fixes on both platforms.
- **CI Screenshot Automation:** Built a **Playwright** runner booting every game headless across device profiles with injected safe-area/notch values, attaching captures to each pull request so reviewers keep a durable visual record after preview links expire; instrumented **~20 games** with **PostHog** analytics.

Tradeindia

Jan 2025 – April 2025

Product & Tech Intern

Noida, India

- **Trust, Safety & Fraud Tooling:** Integrated third-party identity-verification APIs (**Truecaller, IDfy, Shield**) into the buyer-seller flow for fraud detection; drove PoC evaluation of an automated voice bot via transcript analysis and ASR / script tuning.
- **Hybrid Search over Messy Data:** Built retrieval across thousands of unstructured listings combining **FAISS** vector search with FuzzyWuzzy normalization under a weighted semantic-fuzzy-heuristic scoring layer (Streamlit), lifting relevance **89% → 96%**.

PROJECTS

AI Persona - Voice + Chat Agent | *FastAPI, ChromaDB, Voyage AI, Claude, Vapi, Next.js, Cal.com*

- Grounded **RAG** persona over resume + live-fetched GitHub repos serving voice calls and a streaming chat UI (**sub-2s** voice first-token latency via streaming + MMR retrieval); auto-generated “Repo Card” pipeline parses source files into structured summaries, validated by an **LLM-as-judge eval harness** over 20+ adversarial prompts written to break grounding. Live at jiya-persona.vercel.app.

voicequal - Open-Source Audio Quality Library | *Python, PyPI, ONNX Runtime, pytest*

- Audio-quality assessment library shipped to **PyPI** (`pip install voicequal`) – MIT, **75 tests**, published with the full 200-clip benchmark (VocalSet + MUSAN) rather than a favourable subset; ported the decision layer to **Dart** and verified numeric parity to ~9 significant figures against the Python reference, surfacing three genuine port bugs.

Distributed KV-Cache Cluster | *Python, asyncio, Consistent Hashing, Docker Compose*

- Sharded a KV cache across 3 nodes choosing **strong consistency** via synchronous primary-replica replication; transparent **hash(key) % N** request forwarding makes the cluster horizontally scalable without client awareness, with per-node TTL and bounded **LRU** eviction.

EDUCATION

Scaler School of Technology

July 2023 – Present

Undergraduate Program in Computer Science CGR: **7.22**

Bangalore, India

Birla Institute of Technology (BITS)

July 2023 – Present

B.Sc. (Hons.) in Computer Science CGPA: **7.89**

TECHNICAL SKILLS

AI-Assisted Development	Claude / Claude Code, Cursor, prompt engineering, LLM-as-judge evals, RAG, agent orchestration
Languages	Python, JavaScript / TypeScript, Java, Dart, SQL, HTML / CSS
Backend & Infra	FastAPI, asyncio, Firebase Cloud Functions, Node.js / Express, GCP (Cloud Run), Docker, GitHub Actions, Playwright, Vercel, Render
Frontend & Mobile	React, Next.js, Tailwind, Flutter, Phaser
Core CS	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Data & Vector Stores	FAISS, ChromaDB, Firestore, Redis, SQLite, ONNX Runtime

ACHIEVEMENTS

- **LeetCode Knight** – top ~5% by contest rating; competitive programming background.
- **Open Source:** Merged PRs to **NixOS/nixos-summer** (#51) and **OneBusAway/onebusaway-docs** (#140); author of **voicequal** on PyPI.
- **Senior Diploma in Bharatanatyam** – Prayag Sangeet Samiti, Allahabad (rigorous multi-year classical performance certification).